

Oral Soft-Tissue OCT for Clinical Decision Support: Research Directions, Physics, and Hypotheses

Sep 25, 2026 · @Atharva Kocharekar

This document consolidates the discussion so far on building an OCT-based clinical decision support system (CDSS) for oral soft tissue: the anatomy and physics involved, why data scarcity is the defining constraint, the full set of candidate research directions (self-supervised and otherwise), the research questions and hypotheses they lead to, and the decisions still to be made before committing to one.

1. Project context and scope

Goal. Build the research foundation for a clinical decision support system that analyses optical coherence tomography (OCT) scans of **oral soft tissue** — gingiva (gums) and oral mucosa — using OCT alone, and publish the work (journal first, then potentially a conference).

Why soft tissue. Almost all AI work on dental OCT so far concerns hard tissue: caries in extracted teeth, gaps under fillings. Soft tissue is even less explored, which leaves more room for a genuinely new contribution. Soft tissue also connects to high-impact conditions: gum disease, which affects a large share of adults, and oral precancer and cancer, where early detection changes outcomes.

The defining constraint: data scarcity. There is no public dental or oral OCT dataset. Any data will come from the group's own device and collection, likely tens to low hundreds of scans at first, with few abnormal cases. Every direction in this document is chosen with that constraint in mind: the contribution must work, and ideally *shine*, when labelled data is scarce.

Hard tissue as a fallback or extension. The scope is soft tissue, but hard tissue (enamel, dentin, restorations) remains an option if soft-tissue data proves too thin, or as a second evaluation. Section 4 covers what changes if that happens. The recommended strategy is to design a method that is **tissue-agnostic** so the work survives either way.

What this document is for. To lay out every candidate direction in enough detail to choose one, frame a research question and hypothesis, and plan the first paper. It is a decision document, not a final plan.

2. Oral soft tissue: anatomy and what OCT sees

The layers, from the surface down

Oral mucosa has a simple layered architecture, and OCT resolves it well within its ~1-2 mm imaging depth:

1. **Keratin layer (stratum corneum)** — present on *masticatory mucosa* (gingiva and hard palate), absent or thin on *lining mucosa* (cheek, floor of mouth, underside of tongue). On OCT: a thin, bright surface band where present.
2. **Epithelium (stratified squamous)** — layers of tightly packed cells with relatively low scattering. On OCT: a comparatively **dark / low-signal band**. Typical thickness varies strongly by site, roughly from about 100–200 μm (floor of mouth, gingiva) up to several hundred micrometres (cheek). Site must therefore always be recorded, because “thick” in one place is normal in another.
3. **Basement membrane** — the thin interface between epithelium and connective tissue. Not directly visible as a structure, but on OCT the **epithelium-lamina propria boundary** is a sharp, continuous line. Whether that line is intact is the single most important cancer-related feature.
4. **Lamina propria** — connective tissue rich in collagen, blood vessels, and immune cells. On OCT: a **brighter, more strongly scattering band** that fades with depth.
5. **Submucosa** — deeper, looser tissue with fat, glands, and larger vessels; at the limit of OCT depth.

The undulating boundary between epithelium and lamina propria (the *rete ridges*) gives a wavy pattern in healthy tissue; its loss or distortion is informative.

Gingiva specifically

- **Free and attached gingiva** — keratinised tissue around the teeth.
- **Gingival sulcus** — the shallow groove between gum and tooth (normally about 1–3 mm deep). Deepening into a **periodontal pocket** indicates periodontitis.
- **Gingival thickness (phenotype)** — thin vs thick gum tissue; thin phenotype carries higher risk of recession and matters for implants and orthodontics. Currently assessed crudely (visual inspection, probe show-through), so an objective OCT measurement has real value.

Measurable biomarkers

Biomarker	What changes	Linked conditions
Epithelial thickness	Thickening or thinning	Hyperplasia, leukoplakia, atrophy, inflammation
Keratin layer thickness	Hyperkeratosis	Leukoplakia, frictional lesions
Basement-membrane / boundary continuity	Boundary blurs or breaks	Severe dysplasia, invasive carcinoma
Epithelial scattering and structure	Heterogeneity, disordered layering	Dysplasia
Lamina propria attenuation / scattering	Denser or disorganised collagen	Fibrosis (e.g. oral submucous fibrosis), scarring, inflammation
Vessel density (with OCT-A)	More, dilated vessels	Gingivitis, inflammation, tumour angiogenesis
Collagen birefringence (with polarization OCT)	Loss or change of organisation	Fibrosis, cancer invasion
Gingival thickness, sulcus depth	Thin phenotype, pocket formation	Recession risk, periodontitis

Conditions in scope

- **Gingivitis** (reversible inflammation) and **periodontitis** (destructive, with attachment and bone loss).
- **Oral potentially malignant disorders:** leukoplakia (white patch), erythroplakia (red patch), **oral submucous fibrosis** (progressive collagen stiffening, strongly linked to areca-nut use, common in South Asia), oral lichen planus.
- **Epithelial dysplasia** (mild / moderate / severe) and **oral squamous cell carcinoma** — ground truth is biopsy histopathology.
- **Peri-implant mucositis**, and healing after surgery or grafts.

3. OCT physics for oral tissue

Understanding the physics is not background decoration here: several of the research directions depend on it directly.

How the signal is formed (brief recap)

OCT sends broadband near-infrared light into tissue and measures, via interferometry, how much light is back-scattered from each depth. Each **A-scan** is one depth profile; a row of A-scans forms a **B-scan** (cross-section); a stack of B-scans forms a **volume**. The brightness at a depth reflects how strongly the tissue there scatters light back, reduced by everything the light passed through on the way in and out.

Why ~1300 nm is used for oral tissue (both hard and soft)

The choice of centre wavelength is a trade-off between two effects:

- **Scattering falls as wavelength increases.** Tissue scattering decreases roughly as a power of wavelength, so longer wavelengths penetrate deeper through turbid tissue. For enamel, whose tiny crystal structure scatters short wavelengths strongly, scattering at 1300 nm is very low, making enamel almost transparent. For soft tissue, 1300 nm reaches deeper into the lamina propria than 800–900 nm light does.
- **Water absorption rises as wavelength increases.** Water absorbs very little near 800–1050 nm, noticeably more around 1300 nm (on the order of 1 cm^{-1}), and very strongly near 1450 nm.

Over the millimetre-scale paths in oral tissue, water absorption at 1300 nm is modest while the scattering benefit is large, so **~1300 nm is the sweet spot for skin, mucosa, and teeth**. The eye is different: light must cross roughly 2 cm of watery vitreous twice to reach the retina and return, so water absorption at 1300 nm would be prohibitive. That is why retinal OCT uses ~840 nm (spectral-domain) or ~1050 nm (swept-source, a compromise that reaches the choroid).

Practical consequence for this project: oral soft-tissue OCT is physically closest to *other 1300 nm imaging of epithelium over connective tissue* — skin, cervix, oesophagus — not to retinal OCT, even though retinal OCT has far more public data.

Resolution

- **Axial (depth) resolution** is set by the light source, not the lens: $\delta z \approx 0.44 \times \lambda_0^2 / \Delta \lambda$ (in air), where λ_0 is centre wavelength and $\Delta \lambda$ the bandwidth. Because λ_0 is squared, a 1300 nm system needs a much wider bandwidth to match the resolution of an 840 nm system. Example: 1310 nm with 100 nm bandwidth gives about 7.5 μm in air, roughly 5.5 μm in tissue. Typical commercial 1300 nm systems achieve around 5–15 μm in tissue.
- **Lateral resolution** is set by the focusing optics, typically 10–25 μm , and trades off against depth of focus.

- A thin epithelium of ~100 μm therefore spans only on the order of 10–20 pixels in depth, so boundary precision of a few pixels matters clinically.

Refractive index: optical vs physical depth

OCT measures **optical path length**. To get physical thickness, divide by the refractive index: soft tissue is roughly 1.36–1.40, dentin about 1.54, enamel about 1.62–1.63. Any thickness biomarker must apply this correction, and it differs between tissues. Surfaces also refract the beam, so structures below a tilted surface appear shifted.

Attenuation: the core quantitative property

In the simplest (single-scattering) model, OCT signal decays exponentially with depth: $I(z) \propto \exp(-2 \mu_t z)$, where μ_t is the **attenuation coefficient** (mostly scattering in tissue) and the factor 2 reflects the round trip. Fitting μ_t per layer, or estimating it at every pixel with depth-resolved methods, converts a brightness image into a map of a **tissue property**. This is valuable because:

- raw brightness depends on the device, focus position, sensitivity roll-off, and surface reflection; attenuation depends mainly on the tissue;
- epithelium (low scattering) and lamina propria (high scattering) differ clearly in μ_t ;
- fibrosis, inflammation, and dysplasia alter μ_t .

The model has limits: multiple scattering at depth, the focus (confocal) profile, and sensitivity roll-off in spectral-domain systems all distort the simple decay, and must be corrected or modelled.

Speckle

Because OCT uses coherent light, waves scattered from many sub-resolution structures interfere and create a granular pattern called **speckle**. It is usually treated as noise, but its **statistics carry information** about the scatterers: in fully developed speckle the intensity follows an exponential distribution (amplitude follows a Rayleigh distribution), while fewer or more organised scatterers produce different distributions (e.g. K-distribution, Nakagami). Speckle contrast and distribution parameters can therefore act as tissue fingerprints (Section 7).

Polarization

- **Birefringence** — organised fibrous structures (collagen in lamina propria; enamel crystals) delay one polarization relative to the other. Polarization-sensitive OCT (PS-OCT) measures this. Fibrosis and cancer disrupt collagen organisation and change birefringence.
- **Depolarization** — disordered scatterers scramble polarization. Demineralised enamel depolarizes strongly, which is why cross-polarization OCT highlights early caries.

Device-dependent effects to be aware of

Sensitivity roll-off with depth (strong in spectral-domain, weaker in swept-source), focus position, motion (patient breathing, hand-held probe), saliva film causing strong surface reflection, probe pressure compressing soft tissue and changing measured thickness, and the angle of incidence. Soft tissue is **compressible**, so probe contact pressure is a real confounder that barely exists for teeth.

Hard vs soft tissue at a glance

Property	Hard tissue (enamel/dentin)	Soft tissue (mucosa/gingiva)
Structure	Mineralised, rigid; enamel nearly transparent at 1300 nm	Layered epithelium over collagen-rich connective tissue
Refractive index	~1.62 (enamel), ~1.54 (dentin)	~1.36–1.40
Imaging depth	Through full enamel (~1–2 mm) to the dentin–enamel junction	~1–2 mm, limited by scattering
Main contrast mechanism	Scattering increase and depolarization in demineralised enamel	Scattering difference between epithelium and lamina propria; birefringence of collagen
Key features	Lesion depth, DEJ involvement, gaps	Epithelial thickness, boundary integrity, attenuation, vessels
Physical confounders	Surface reflection, refraction at enamel surface	Compression, motion, saliva, site-dependent anatomy
Ground truth	Micro-CT, microradiography (ex vivo)	Histopathology (biopsy), clinical assessment

Property	Hard tissue (enamel/dentin)	Soft tissue (mucosa/gingiva)
Ease of data collection	Easier: extracted teeth, ex vivo	Harder: living patients, ethics, biopsies
Closest public source domains	Few; mainly simulation	Skin, cervical, oesophageal OCT (1300 nm, epithelium)

4. The hard-tissue angle (if the project switches or expands)

If soft-tissue data proves too thin, or a second evaluation is wanted, hard tissue is the natural extension. The methods in Sections 6–7 are largely tissue-agnostic, but several specifics change.

Targets. Early (non-cavitated) caries and its depth relative to the dentin–enamel junction (DEJ); gaps and debonding under composite fillings; secondary caries at filling margins; enamel cracks; erosion; plaque on the tooth surface.

Physics that matters. At 1300 nm sound enamel scatters very little, so the DEJ is visible beneath it. Demineralisation creates micro-pores that increase scattering and **depolarize** light; cross-polarization or PS-OCT therefore shows early lesions with high contrast. Lesion severity is often summarised by **integrated reflectivity** (ΔR) over lesion depth. Enamel's high refractive index (~ 1.62) means optical depth overstates physical depth by $\sim 60\%$, and the sloped tooth surface refracts the beam.

Data. Considerably easier. Extracted teeth are plentiful from clinics, need minimal ethics compared with living patients, can be imaged repeatedly under controlled conditions, and can be validated against **micro-CT** or transverse microradiography, which give precise mineral-loss ground truth. Artificial lesions can be created in the lab (demineralising solutions) to produce graded, controlled examples — effectively a physical data generator.

Source domains for transfer. Enamel at 1300 nm is not layered like mucosa, so skin or cervical OCT is no longer a close match. Transfer would come mainly from unlabelled dental OCT, simulation, and lab-generated lesions.

Main caveat. Most hard-tissue results are ex vivo. Moving into the mouth adds saliva, motion, and restricted probe access (posterior and between-teeth surfaces are hard to reach) — an ex-vivo-to-in-vivo domain shift that is itself a research problem.

Recommended hedge. Frame the method as *physics-grounded, label-efficient learning for oral OCT*, evaluate primarily on soft tissue, and keep hard tissue as an optional second evaluation. That protects the paper if either data source disappoints, and a method shown on two very different tissues is stronger evidence of generality.

5. Why data scarcity shapes the research

Data scarcity is not just an obstacle to work around; it defines what counts as a contribution.

What scarcity looks like here.

- **Few scans overall** — likely tens to low hundreds of patients at first.
- **Very few abnormal cases** — healthy mucosa is easy to scan; dysplasia and cancer are rare and need biopsy confirmation.
- **Expensive labels** — pixel-level layer masks take expert time; diagnostic labels need histopathology.
- **Single device, single site** — so generalisation cannot be assumed.
- **Site heterogeneity** — gingiva, cheek, tongue, and palate look different even when healthy.

Consequences for method design.

1. Standard supervised deep learning from scratch will overfit; reported numbers would not be trusted.
2. A contribution should **extract more from less**: unlabelled data, physics, anatomical priors, related domains, or cheaper labels.
3. Evaluation must be **honest at small scale**: patient-level splits, repeated cross-validation, confidence intervals, and results reported across shrinking label budgets.
4. Outputs should be **measurements with uncertainty**, which are both more useful for decision support and easier to validate than a single diagnosis label.

The central framing. The paper's value should rest on the claim “*this approach works when labelled oral OCT data is scarce, and here is why,*” rather than on a headline accuracy number. That makes scarcity the research question rather than a limitation paragraph.